

POINT HOPE, AK, USA

WMO: 701043

Lat: 68.350N

Lon: 166.800W

Elev: 4

StdP: 101.28

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26623

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.4	(c) -29.8	(d) -34.0	(e) 0.2	(f) -26.0	(g) -33.6	(h) 0.2	(i) -27.1	(j) 21.0	(k) -18.0	(l) 19.1	(m) -16.8	(n) 4.4	(o) 320	(p) 1.089

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 3.2	(c) 14.9	(d) 12.0	(e) 12.8	(f) 10.7	(g) 12.1	(h) 10.1	(i) 12.4	(j) 14.4	(k) 11.1	(l) 12.8	(m) 10.1	(n) 11.6	(o) 4.6	(p) 10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.3	8.3	13.7	9.9	7.6	12.2	8.8	7.0	11.2	35.5	14.4	31.8	12.8	29.0	11.5	15.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 17.9	(b) 15.8	(c) 14.3	(e) -35.4	(f) 17.9	(g) 3.6	(h) 2.8	(i) -38.0	(j) 19.8	(k) -40.2	(l) 21.4	(m) -42.2	(n) 23.0	(o) -44.8	(p) 25.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.6	-18.4	-18.3	-16.9	-11.3	-1.7	3.5	7.9	8.0	4.9	-2.1	-8.7	-15.5		
	DBStd	11.26	7.36	7.39	7.19	6.20	4.01	3.27	2.90	2.32	3.21	4.58	5.32	6.43		
	HDD10.0	5734	882	792	835	639	362	197	78	70	154	374	562	791		
	HDD18.3	8749	1140	1025	1093	889	620	444	322	320	403	632	812	1049		
	CDD10.0	27	0	0	0	0	0	3	13	9	2	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	6.8	7.0	6.9	6.7	6.1	5.5	5.6	6.0	6.7	7.2	7.9	8.2	8.3		
(15) Precipitation	PrecAvg	462	30	31	19	19	18	28	57	79	53	58	41	29		
	PrecMax	627	76	77	57	53	44	50	98	149	90	98	116	58		
	PrecMin	314	3	3	3	3	0	12	18	18	11	19	7	3		
	PrecStd	78	18	20	15	14	11	9	19	32	19	20	26	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.0	0.2	1.1	2.7	7.6	16.0	17.9	16.4	12.8	9.1	2.8	0.0		
		MCWB	0.4	-0.3	-0.2	1.0	2.9	11.0	13.2	13.5	9.9	5.8	2.2	-0.5		
	2%	DB	-1.8	-1.3	-1.0	1.4	5.2	12.4	16.1	13.8	11.5	6.2	2.0	-1.2		
		MCWB	-2.3	-1.4	-1.5	-0.1	1.9	8.8	12.7	11.6	9.5	4.6	1.3	-1.6		
	5%	DB	-3.9	-3.0	-2.8	0.2	3.5	10.4	13.6	12.5	10.6	4.7	0.4	-2.7		
		MCWB	-4.8	-3.5	-3.6	-1.5	0.7	7.9	11.4	10.8	8.8	3.5	-0.4	-3.2		
	10%	DB	-7.2	-6.5	-5.6	-2.0	2.6	8.1	12.3	11.3	9.1	3.3	-1.3	-5.9		
		MCWB	-7.8	-6.9	-6.3	-3.0	0.1	6.0	10.4	9.8	7.8	2.2	-1.9	-6.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.3	-0.3	0.1	1.5	5.0	11.8	14.3	14.0	10.7	6.5	2.4	-0.5		
		MCDB	0.8	-0.1	0.9	2.4	6.6	14.6	16.9	16.0	12.0	8.6	2.6	-0.3		
	2%	WB	-2.2	-1.5	-1.5	0.5	3.0	9.9	13.0	12.1	9.9	4.9	0.9	-1.7		
		MCDB	-1.8	-1.4	-1.1	1.3	4.1	12.1	15.4	13.4	11.2	6.1	1.7	-1.4		
	5%	WB	-4.5	-3.4	-3.5	-1.0	2.0	8.0	11.7	10.9	8.9	3.6	-0.4	-3.2		
		MCDB	-3.6	-3.1	-2.8	-0.3	2.9	9.6	13.5	12.4	10.1	4.6	0.3	-2.5		
	10%	WB	-7.4	-6.9	-6.1	-3.0	1.2	6.5	10.3	9.9	8.0	2.5	-2.1	-6.0		
		MCDB	-6.9	-6.5	-5.4	-2.2	2.1	8.1	12.0	11.2	8.9	3.4	-1.6	-5.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	5.9	6.1	6.4	4.4	3.9	3.9	3.2	3.3	3.9	5.0	5.1		
		MCDBR	7.6	7.5	5.8	6.9	5.6	7.0	6.6	4.6	3.9	3.6	4.2	5.9		
	5% WB	MCWBR	6.9	7.4	5.6	6.2	4.2	4.7	4.4	3.4	2.7	3.3	3.9	5.9		
		MCWBR	7.1	7.6	5.6	6.7	4.8	6.2	6.1	4.4	3.4	3.3	4.2	5.5		
(40) Clear-Sky Solar Irradiance	taub		0.160	0.224	0.253	0.323	0.354	0.343	0.363	0.371	0.311	0.260	0.358	N/A		
	taud		1.899	2.049	2.171	1.989	2.029	2.398	2.450	2.374	2.505	2.232	2.821	N/A		
	Ebn at Noon		114	571	789	802	821	857	824	764	738	522	148	0		
	Edh at Noon		7	46	77	127	137	99	92	88	59	39	12	0		
(44) All-Sky Solar Radiation	RadAvg		0.03	0.51	2.23	4.44	5.62	5.94	4.65	3.17	1.81	0.56	0.07	0.00		
	RadStd		0.00	0.06	0.16	0.24	0.58	0.45	0.42	0.39	0.17	0.08	0.01	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page